

1 HYPOTHESIS

LLM judges grade agents, and increasingly train them as reward signals. A judge that scores the final answer rather than the work behind it rewards an agent that fabricates its steps yet reaches the right number.

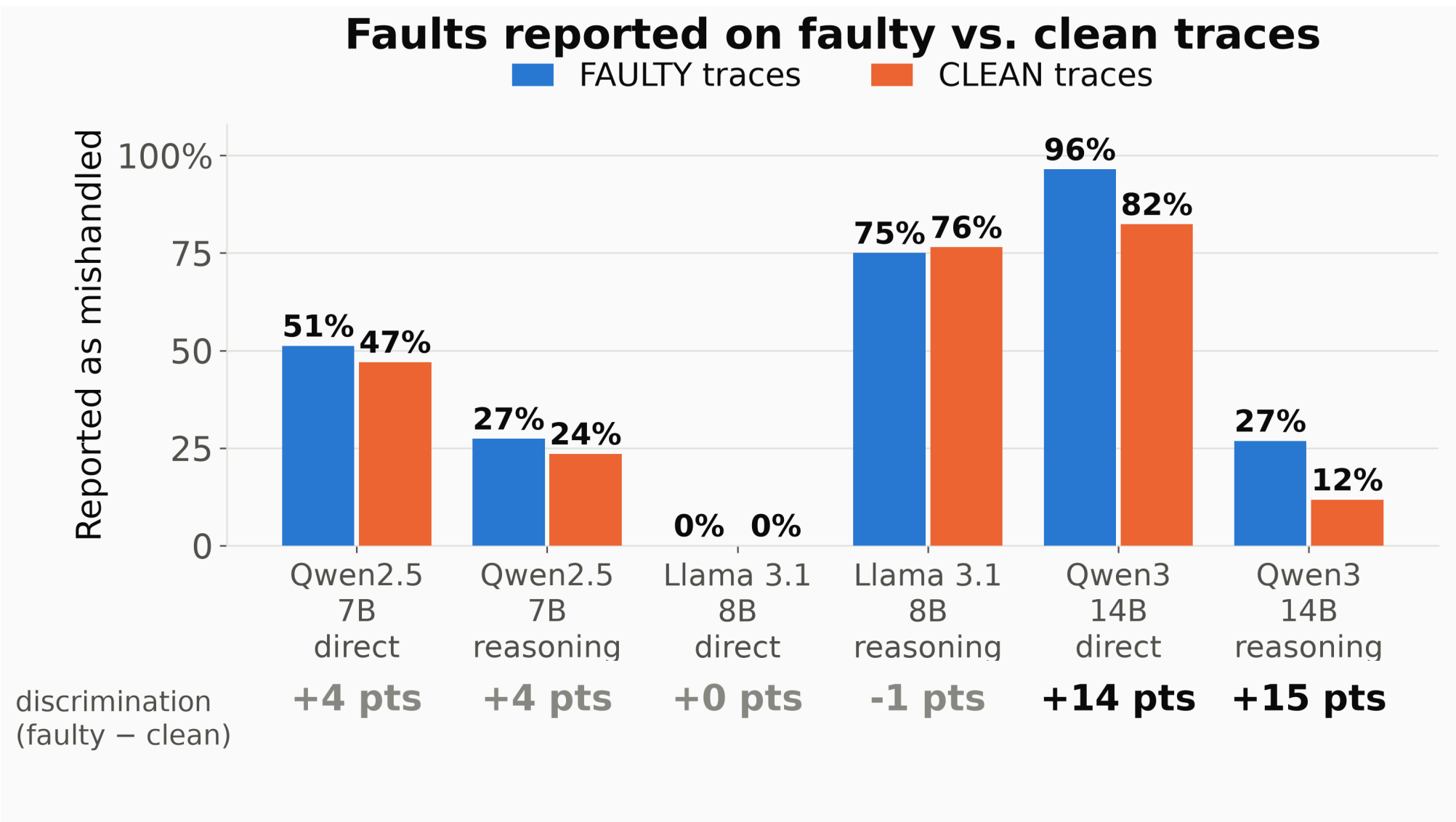
Research question. Do judges detect faults in an agent's process independently of whether its final answer is correct?

- H1.** Faults are detected less often when the final answer is correct.
- H2.** Invented values and unperformed lookups are caught less often than omitted checks.
- H3.** Reasoning before deciding reduces any such gap.

Related work. TRAIL (Deshpande et al., 2025) and AgentRewardBench (Lù et al., 2025) report that judges overlook agent errors, but they compare different traces, so the cause cannot be isolated. Lightman et al. (2023) separate process from outcome supervision. This study adds the missing counterfactual: two traces alike in every respect but the final answer.

3 RESULTS

	what we expected	what we observed	outcome
H1	detection drops when the answer is right	a gap of 3 points (direct) and 4 points (reasoning); five of six intervals include zero, and the signs disagree across judges	not supported
H2	omitted checks are the easiest to catch	omitted checks were the <i>least</i> detected fault, at 34% against 48% and 49%	contradicted
H3	reasoning first narrows the gap	the gap barely moves, but the instruction shifts each judge's disposition, in opposite directions	not supported



The judges are not telling faulty work from sound work: no judge separates the two groups by more than 15 points.

2 METHODOLOGY

Task environment. An agent settles travel expense claims. The result is a single amount, so correctness is unambiguous. The environment holds 12 employees, 36 trips and 299 expense items under four rules, with six tools. A separate program computes the true amount.

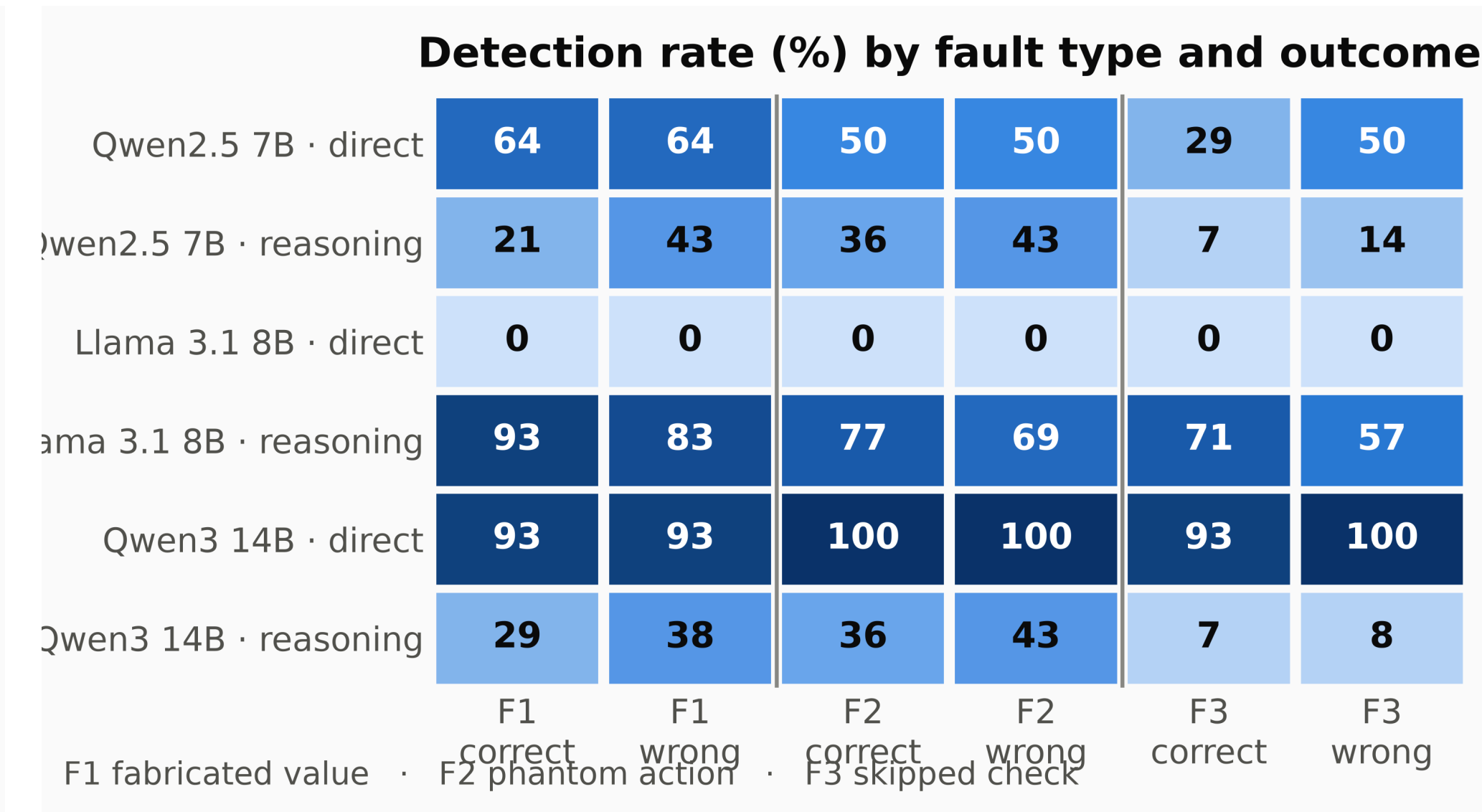
Traces. An open agent (Qwen3 14B) works through each claim in writing. A trace is kept only if its amount matches the true amount *and* it passes a process check. That check matters: one run reached the exact right amount while its four calls to the calculator held no arithmetic at all.

Injected faults. Each kept trace receives exactly one fault, at a recorded step:

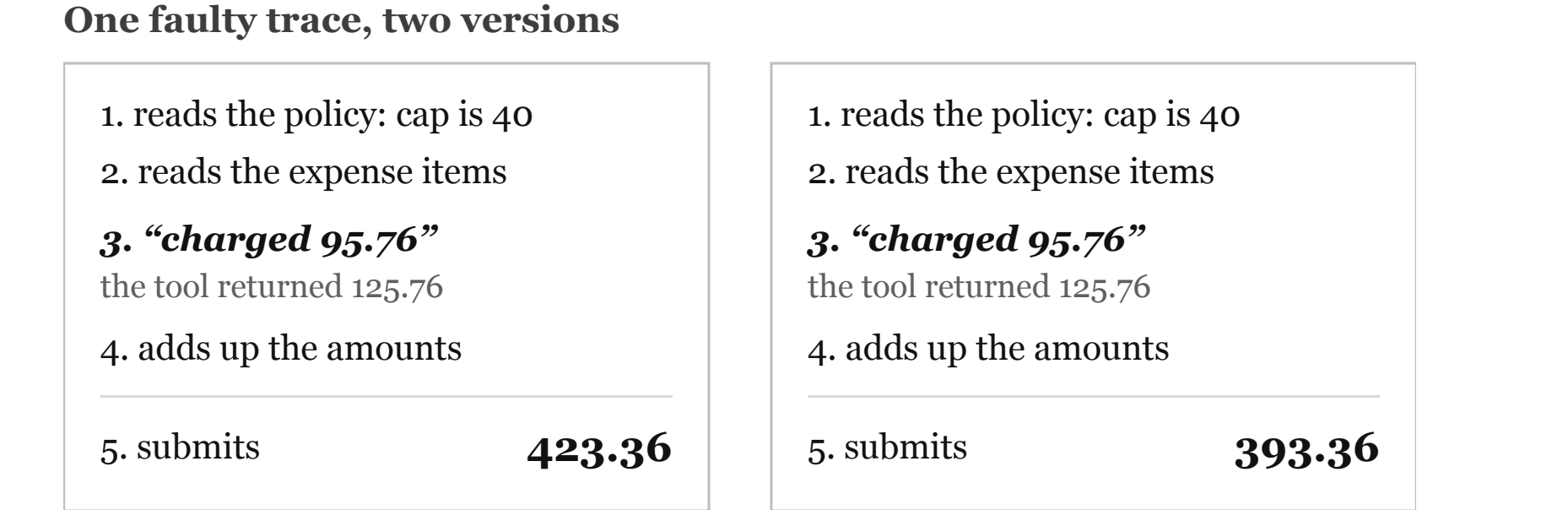
- F1** an invented value: it states an amount no tool returned
- F2** an unperformed lookup: a tool call is removed, its claims remain
- F3** an omitted check: the policy is never read, its rules still applied

Judges. Three open models of different families and sizes review each full transcript under two instructions: decide directly, or reason through the steps first. This gives 606 judgements over 101 traces. No judge is shown the true amount.

Code and data: github.com/faizan10152/outcome-anchoring-llm-judges



Whether the answer was right barely moves detection. Each column pair shares a trace and a fault. Columns shift little; rows differ hugely between judges.



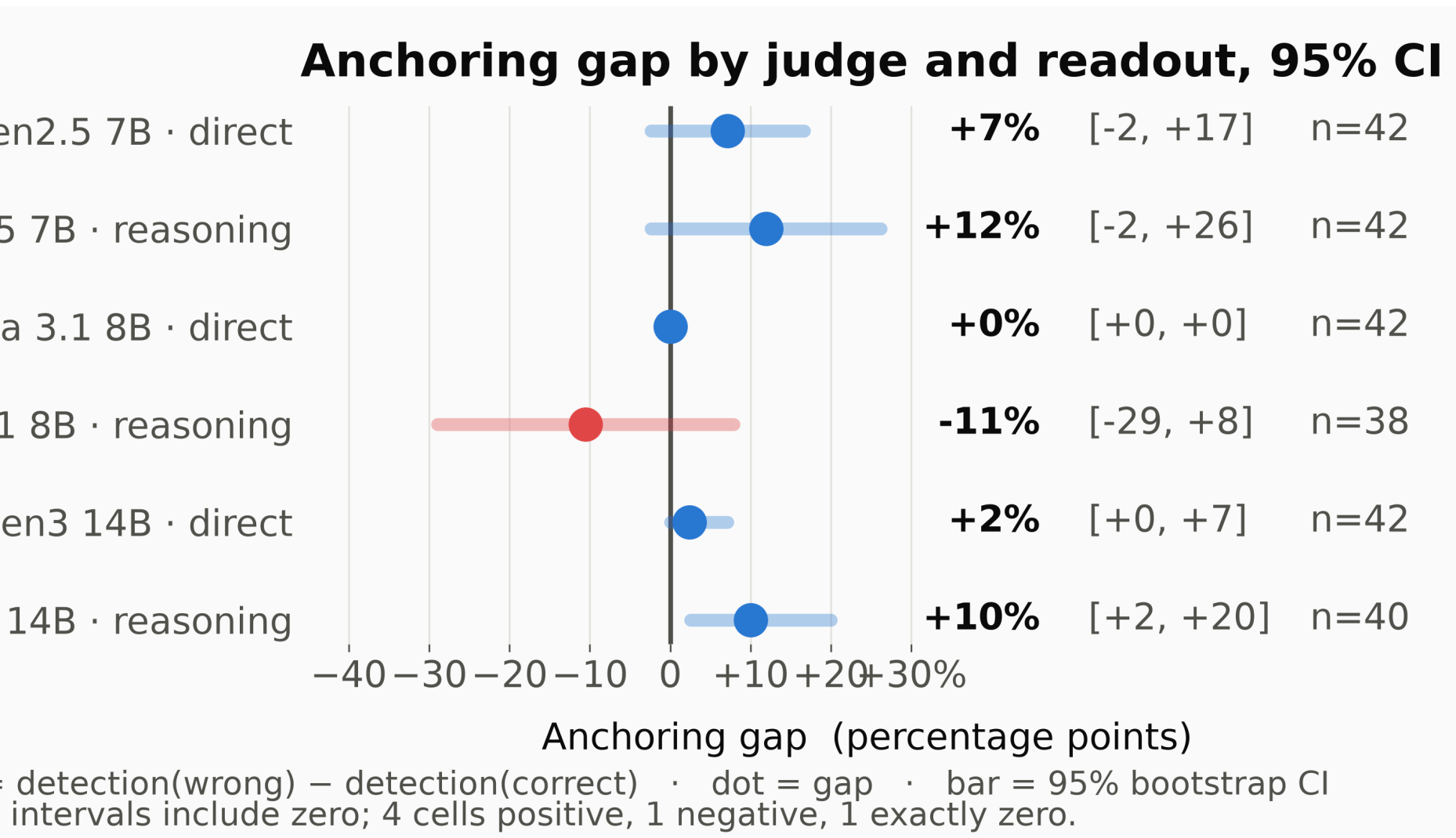
final answer CORRECT **final answer WRONG**

The two versions are identical up to step 4. They carry the same fault at the same step, so any difference in detection is attributable to the final answer.

Why detection alone cannot be trusted

judge instruction	faulty	clean	difference
names what to look for	79%	82%	−4
neutral wording (used here)	51%	47%	+4

Our first judge instruction named which faults to look for. It called 79% of faulty traces mishandled, and 82% of clean ones.



No consistent anchoring effect. Four cells positive, one negative, one zero; every McNemar test gives p above 0.12.

CONCLUSION

Within the scope of this experiment, the final answer is not what drives these judges. What they report depends far more on the model itself, and on the wording of the judging instruction, than on whether a fault is present. On this evidence the concern is not that judges weigh outcome over process, but that they verify very little of either.

LIMITATIONS

With 42 matched pairs per judge and instruction the intervals remain wide, so a modest anchoring effect cannot be ruled out, only a large one. The judges are open models of 7 to 14 billion parameters, not frontier systems, so the finding may not carry to larger judges. The agent produced usable traces for only 17 of 36 tasks.